



"Water Monitoring in Indian Communities by Women: A Research Summary"

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summary

This report is the final output of the project "Monitoring Water Conditions in Indian Communities by Individuals." The project is affiliated with the Nitzan Laboratory at Tel Aviv University, under the guidance of Prof. Selda Adris.

In India, as in many developing countries, there is a severe issue of drinking water pollution. The project's goal is to provide a solution for monitoring water quality in Indian communities, empowering women to independently assess the water they use in their households. This initiative aims to address health problems and epidemics caused by polluted water, ultimately improving the quality of life for residents.

The project encompassed an overview of various water pollutants, with a specific focus on Karla, the region in India where the project is centered. Additionally, we conducted in-depth research on existing household measurement tools available in the market, analyzing their suitability for the project based on parameters such as cost, usability, environmental impact, reliability, and more. During the project, we sought similar studies conducted in the past in developing countries, which were based on community-led household water monitoring. Most of these studies relied on laboratory tests conducted by water experts, providing us with a deeper understanding of the complexity of the issue.

Reviewing articles related to similar monitoring programs helps to identify the most common pollutants. Analyzing these findings allowed us to identify the type of pollutant, the testing method and scope, the location, and the timing of testing. This data assisted us in selecting the most critical pollutants, depending on the impact of their entry into the body. Another crucial parameter in selecting these pollutants was the possibility of simple measurement using readily available tools for the Indian population.

This work is preliminary, laying the foundation for further project development by additional teams. It is recommended that future teams integrate the concept of household measurement with an application containing a basic water quality survey capable of storing data over time, enabling the identification of similar issues in the same geographical area.

Introduction

Access to clean water and the ability to monitor water quality are fundamental conditions for sustaining and managing life. Water scarcity is one of the major challenges globally, particularly in developing countries like India. The essential and ongoing demand for drinking water necessitates the use of various technological mechanisms for sustainable water resource management. Factors negatively affecting water resources include population growth, climate change, improved living standards, water resource shortages, and territorial size. To accelerate addressing the water issue, the United Nations declared 2018-2028 as the International Decade for Action "Water for Sustainable Development" (Government of Kerala, 2019).

In developing countries¹, approximately 2.4 billion people, a third of the world's population, face a scarcity of water, and many of them suffer from perpetual water shortages. Their daily routines involve irregular water supply that does not meet minimum drinking water standards, leading to unsanitary conditions (Hamidreza & Zohreh, 2012). The water crisis adversely affects all aspects of life in developing nations, with the most severe impact evident in rural communities. Two-thirds of the global rural population suffer from inadequate sanitation conditions defined by the WHO², such as lack of access to safe water and proper sanitation facilities. This issue also affects 14% of urban dwellers worldwide. In Asia, 693 million people lack secure access to clean water, with 86% of them living in rural areas. In Africa, 300 million people lack access, with 85% of them in rural regions. In South America and the Caribbean, this number is 78 million (Kathuria, 2006).

Many developing countries are located in water-scarce geographical regions. However, there are also countries with abundant water resources where economic conditions prevent the development of efficient water distribution systems. For example, in Asia, the continent richest in water, the number of people suffering from proper drinking water scarcity is second only to the arid regions of Africa. Often, this is due to short and intense rainfall seasons, requiring water storage methods that are absent in most countries. In India, rainfall is primarily concentrated in the monsoon season, providing a significant portion of the annual precipitation within a few hours. However, monsoon water is not necessarily suitable for most household uses.

Additionally, inefficient separation between water and sewage systems leads to a lack of clean water access for a significant portion of the population, impacting public health and life expectancy

¹ "A country in which there is a low level of material prosperity for its residents, such as infrastructure, technology, education, economy, health, and services."

² World Health Organization

(Kathuria, 2006). If so, there is no guarantee of water quality, and it's challenging to determine if it's suitable for use. This problem leads to the need for water quality indicators. A biological test must be conducted within a 6-hour range from sample collection. In developed countries, this process is simple and accessible. For example, in Israel, anyone can request a water quality test, either by phone or online (see Appendix 1). However, in developing countries, the situation is different, and such accessibility is often nonexistent. Additionally, not every family can afford to finance these tests, certainly not at the required frequency (Abdi & Delavari, 2018). This creates a fertile ground for pollution and diseases with short and long-term effects.

Research indicates that the presence of sanitation facilities in schools influences girls' attendance: when school facilities provide adequate sanitation conditions and privacy in restrooms, there is an 11% increase in girls' attendance on average (Hamidreza & Zohreh, 2012). Furthermore, in many developing countries, a patriarchal culture persists. Children, along with women and girls, bear the traditional role of fetching water from the nearest source, often leading to a Sisyphean daily task. A crucial aspect of this project is women's empowerment and creating equality by emphasizing the essential role of women in the community. The idea is for women to monitor water quality and take responsibility for ongoing measurements and operations. The project aims to provide an individual-level solution for water monitoring, meaning for each household. The aspiration is that water quality testing for each family, under female responsibility, will improve overall community quality of life (Trent et al., 2018).

In terms of SDG³s, the project aims to address Goal 6, Clean Water and Sanitation. The objective is to ensure the sustainable availability and management of water and sanitation for all. More specifically, the focus is on Target 6.B, support local engagement in water and sanitation management, support and strengthen the participation of local communities in improving water and sanitation management (Sustainable Development Goal Website). Consumption of polluted water or the absence of proper sanitation conditions leads to gastrointestinal diseases, malaria, dysentery, infections caused by bacteria, as well as severe and prolonged diarrhea, which may result in death from dehydration. Patients with such diseases, unable to wash their hands with water and soap, pose a serious risk of infection to their environment. The education of children suffering from chronic diseases (often originating from water), such as parasites, is affected: even if they manage to go to school, they are weak and lack the concentration needed for learning (Abdi & Delavari, 2018).

³ Sustainable Development Goals

Methodology and Approach to Solution

The project aims to provide a solution for monitoring water quality in communities in India. The general idea of the proposed solution consists of two parts: responding to a survey and measuring water samples using household measuring tools. These data will be sent to the research institute for collection and analysis. Since we are in the initial stage of the project, we focused on the drinking water sampling part. Proposals for the continuation of the project and the survey will be discussed in the next section. The theory of change is divided into three channels:

1. **Increasing Awareness** - We believe that by increasing awareness, we can change daily water practices and thus partially solve the issue of pollution. Raising the topic of water pollution to the agenda of the people in the destination country will lead to a desire for change (Trent et al., 2018). In fact, change can only happen when participants feel the need for it. Many people are not aware of the impact of water pollution, even though they experience it daily. When they become aware of the connection between their illnesses and the quality of the water they consume, they will adopt the understanding of how water quality affects their quality of life, be open to options for change, and agree to take the required responsibility. Increasing awareness along with proper tools and guidelines (hygiene principles, transmission, and storage) have proven to be effective in improving water conditions (Okoyere, Pangaribowo, Asante, & Braun, 2017).
2. **Community Building** - Creating a community is a social and personal solution. It involves collective coping with problems that an individual alone cannot handle. While there is no guarantee that collective effort will succeed where individuals fail, the very partnership, involvement, and commitment of people to achieve a common goal, influence decisions affecting their lives, and improve their quality of life in their environment, create a new sense and new capabilities among the participants - and this is an important outcome in and of itself. The community provides its members with essential needs that people who live without them and lack a sense of community are not aware of. Alienation can become existential unless a person feels a sense of belonging to a body in which mutual trust and commitment to shared goals exist (Sadan, 2002). The project is expected to take place in Kerala, and the aspiration is that it will happen in several households within the same community. Connecting multiple families to the project will create a supportive community that will sustain its continuity. The mere knowledge that one is part of something bigger preserves their activism. The participation of several members from the same community can change the quality of life for

the entire community, which can encourage them to engage in the project and even continue to participate, driven by a sense of community commitment.

The market is indeed flooded with water quality monitoring devices, but these devices are expensive, technologically complex, and consume a lot of electricity. Our approach requires an affordable and practical solution suitable for the specific target audience of women who do not come from a technological background and may not cope well with complicated technology. We want to increase their awareness regarding 'proper' water usage: they should be able to determine, based on instructions, whether the water is suitable for drinking or not, and also understand the consequences of consuming water with certain chemicals and particles. Throughout the sustainability and continuity of the project, the proposed solutions have been ecologically focused and are intended for repeated use. By continuously raising awareness about the water situation, we will succeed in influencing residents to increase their complaints to the authorities. The overall aspiration of the project is to have a long-term impact.

One of the analytical methods for assessing water quality today is the Water Quality Index (WQI), a mathematical tool used to convert large quantities of water characterization data into a single number that represents the level of water quality. Using this method, an index can be obtained for each water source based on data obtained from tests. The water quality index provides a value based on various parameters such as pH, Conductivity, BOD, COD, Hardness, TSS, Sulfate, Chloride, Turbidity, Nitrates, and Coliform. The weighted arithmetic WQI combines information from the selected water quality parameters into a mathematical equation and provides a single number that quantifies the impact of all parameters together (Anoop, Arathy, Markose, Merin, & Riya, 2019). Based on the survey questions and the WQI equation, it will be possible to arrive at a number that provides a brief explanation of the water condition. Its accuracy will not fully reflect the overall situation but will give an indication of whether the water is suitable for drinking or not. To increase the likelihood of project success, we relied on similar research conducted in the field to understand how to deal with the problem.

Literature Review

Our research is based on the perception that creating a database and disseminating knowledge can bring about change. This perception is grounded in numerous studies. One of the studies collected data from 60 rural and urban settlements with populations ranging from 100 to 1000 households in the Kanpur district of Outer Pradesh, India (see Appendix 5, Table 4). The research used laboratory tests as well as a simple and affordable field test. The study tested *E. coli* using the EPA Method 1604 and also used the CONTAMINATION SIGNAL for separate testing. Additionally, the USEP Method 1604 was used to test the quantity of coliforms and *E. coli* in water using MI Medium. The home tests were conducted using the Open-Source Aquatest (see Appendix 5, Table 4).

The research divided the participant groups into 3 categories: the control group, which only received information for proper water usage and preservation; the standard test group, which received information for proper water usage and preservation, and their findings were tested for quality, and the participants were informed of the results; and the "kits" group, in which each household was provided with ten water testing kits, yielding semi-quantitative results for *E. coli*. The participants were encouraged to use these kits as they saw fit. The study included ongoing education and community engagement through meetings with homeowners, visits to improve water management, and instructions on hygienic water behaviors. The study also included self-reporting by participants on topics such as water usage frequency, water treatment methods, source of drinking water, water storage practices (e.g., covered storage containers), and availability of soap for handwashing. Additionally, the researchers conducted home visits twice, initially and four weeks later when surprise water samples were collected for *E. coli* testing.

The researchers aimed to demonstrate that community education on water management and preservation would improve the quality (as evidenced by the number of *E. coli* findings) of the water. Furthermore, home education on managing domestic water and providing specific water quality information to families would influence their water behaviors (storage, treatment, and maintenance) and thereby improve the quality. The article presents a positive possibility that sharing information on water preservation as an indicator can lead to better water preservation and quality (see research results in Appendix 5, Table 5). The findings indicated that providing information on water quality and preservation methods had short-term positive effects. Notably, significant differences were observed when inappropriate water quality was identified (e.g., the kit group saw an 11% increase in water treatment compared to other groups) (Trent et al., 2018).

In another experiment conducted in Ghana, it was found that providing information and domestic water testing were effective in promoting safe water management behaviors. Community members were randomly selected to participate in group workshops tailored for adults or school children and were subsequently given water testing kits (Acquagenx's Compartment Bag Test (CBT)) to use as they saw fit (differing from the previous study, which also included home visits). There was relatively high demand for the water testing kits, with approximately 50% of recruited adults and 79% of recruited children choosing to participate in the workshops. Both treatment groups showed improvements in safe water management behaviors compared to the control group, which received no information or testing equipment (Okyere et al., 2017).

For more on the positive impact of providing information on water quality, you can refer to the article by Davis et al. (2011). As part of our work, we gathered information from various studies that examined water pollution in general and in Karla specifically, with the aim of creating a suitable WQI (Water Quality Index). For a detailed summary of the studies examined and their findings, please refer to Appendix 2. One study assessed the quality of tap water in 14 districts in Karla. Various parameters were tested. The results showed that all water samples were contaminated by Coliform Bacteria, with 15-20% exceeding the standard. Parameters such as pH, fluoride, and TDS exceeded the standard by 12.8%, 8.1%, and 10.5%, respectively. Another finding revealed that drinking water supplied in rural areas was less polluted than water supplied in urban areas due to improper management of solid and liquid waste. Different tests were performed for various parameters, including pH, water temperature, TDS, salinity, total suspended salts, and nitrates, using Extech Combo electrodes and Orions selective electrodes. More significant contamination tests such as chloride, hardness, magnesium, calcium, sodium, potassium, fluoride, sulfate, phosphates, and coliform bacteria were conducted in the laboratory (Karthick et al., 2015).

Another study conducted in Kerala assessed the groundwater conditions and the quality of domestic wells using a geographical information system. The study involved 449 households. Data collection was carried out through a structured questionnaire and the Differential Global Positioning System (GPS), while water samples were analyzed in a laboratory. The results revealed that in the majority of the domestic wells, the water was highly contaminated beyond acceptable levels. For example, coliform contamination was found in 73% of the samples, and

groundwater contained acids, rendering it unsuitable for drinking. The researchers' assessment was that proper cleaning and chlorination⁴ were lacking.

In water wells, they caused widespread coliform contamination (Ananth et al., 2018). Another study conducted in Karla examined the waters of two main rivers - Priyar and Barathoza. The research found that the primary causes of water pollution stem from anthropogenic activities⁵)Anoop, Arathy, Markose, Merin & , Riya , 2019(

In parallel with the creation of a suitable WQI for the project, we attempted to find similar research or projects. A comprehensive literature review (see Appendix 4) revealed that most studies utilized laboratory measurement methods rather than on-site testing. Typically, the tests were conducted using sophisticated equipment with high costs. For instance, one study examined the water quality of the 'Piyar' river in Kerala, South India (see Table 1 in Appendix 5). The parameters tested included pH, conductivity, Total dissolved solids, hardness, Dissolved Oxygen, Chemical Oxygen Demand, Calcium, Magnesium, Sulphates, Chlorides, Nitrate-N, Iron, and Fluoride. However, among all the pollutants tested, only pH, conductivity, and Total dissolved solids were tested on-site using expensive digital equipment (Lakshmi & Madhu, 2014). Therefore, we needed to find studies that utilized on-site measurement methods with low financial costs. The most common method in this category is testing based on hydrogen sulfide (H₂ S). Biological pollutants such as the bacteria *Citrobacter freundii*, *Salmonella* spp., and *E. coli* produce H₂ S (Allen, Taylor, & Geldreich, 1980). For this reason, the H₂ S presence detection test can be used to identify biological pollutants. One of the studies that demonstrated this possibility originated from the outbreak of cholera in the city of Gwalior, India. The researchers were tasked with finding on-site water monitoring methods. The team utilized a low-cost field test to detect pollution through H₂ S. They compared the results to the standard laboratory testing... (end of text)⁶ MPN , This is a reliable test but is complex to perform and time-consuming. The new test was rated as preferable compared to the MPN, except for the difficulty in accurately determining the number of coliforms (Manja, Maurya, & Rao, 1982).

Another study conducted using a similar method, in Chile, collected 622 samples of drinking water from three regions in the country, under both 'treated' and 'untreated' conditions. The samples were tested using the H₂ S strip method and the MPN method, which includes coliform bacteria

⁴ Water disinfection using compounds containing active chlorine, with high oxidation capabilities, capable of killing bacteria and algae.

⁵ Human activities' impacts on nature, such as industrial pollution, deforestation, and more.

⁶ Most Probable Number (MPN) - A test used to estimate the concentration of living microorganisms present in a water sample.

and coliphages. In this study, sodium thiosulfate, total coliform densities, coliphage densities, pH, residual free chlorine (DPD) were also tested. The H₂S field method was compared to laboratory methods. MPN and APHA⁷. The research results showed that the tests were equally effective in predicting the presence of bacterial contamination in drinking water, except for a difference in the number of tests, which seems to stem from the sensitivity of the MPN test, which may not necessarily be preserved under laboratory conditions. The research results confirm that the H₂S strip test is a sensitive microbiological process for water testing, suitable for data validation (see Table 2 in Appendix 5) (Castillo, et al., 1994).

Monitoring and assessment.

Many households in India use water stored in various containers, and their quality depends not only on the pollutants present in the water source but also on the time spent in the containers and the sanitary conditions. Due to this, there is a need for frequent water quality monitoring. Our project will include a survey aimed at characterizing specific pollutants and tracking water consumption habits in India, specifically in Kerala. Additionally, each household will conduct measurements using a designated device or stick. These two monitoring channels will provide a comprehensive picture that will enable us to devise tailored solutions to address water pollution in Kerala. In the future, we aim to link the collected data with Indian authorities so that long-term pollutants in the same community can be identified and addressed effectively.

The data collected from the survey and measurement results will be collected and monitored by the joint team from Amrita University and the Nitzan lab. In the current project stage, the main goal is to understand the current situation on the ground. The team will continuously receive and analyze the data to identify trends related to the water source, seasons, storage methods, and more. The findings from this initial learning phase will inform the development of a humanitarian plan that addresses specific issues in each community and assists in water purification to provide clean and suitable drinking water. Due to economic and technological constraints, monitoring and cleaning water from all pollutants is challenging, thus the project will focus on specific pollutants. Water pollutants can be divided into those causing immediate harm and those causing long-term harm. The immediate harm pollutants will take precedence over those with long-term effects.

The cost-benefit analysis of the project will be based on what is required from the people in Kerala in terms of investment, compared to what they will gain from participating in the project. The

⁷The Animal and Plant Health Agency (APHA)

people in Kerala will need to invest one-time financial costs, such as purchasing monitoring devices, or periodic costs for consumable products like the sticks, given their disposable nature. Additionally, there will be a mental cost of investing time and effort in consistent water monitoring, which will require increased attention. From this perspective, the financial expense that the users incur will eventually return to them, and they may even gain economically in the long term as they will be able to work more days. Cost-benefit analysis specific to each potential measurement device will focus on several parameters:

- Device availability in developing countries: To ensure the project's sustainability, the device must be locally available, as importing testing equipment and its components over time is unrealistic.
- Financial cost of the device: Given that a significant portion of the population in third-world countries is impoverished, they won't be able to invest high sums in purchasing measurement devices. We're targeting areas where the average person earns up to \$1,000 per year (the bottom of the pyramid), so it's essential to choose an affordable option that users can afford.
- Severity of the damage caused by the unacknowledged pollutant: Since we can't test for all existing pollutants, it's essential to select the most critical ones that could potentially lead to significant harm if consumed unknowingly.
- Test reliability: Results from field tests, even if conducted in a lab setting, may not always be accurate enough to rule out the presence of a pollutant. It's important to choose a device that provides reliable results.
- Ease of use: Water quality testing is not straightforward, especially in field conditions. There's a need for suitable tools, accurate water measurements, proper storage, and more. Extensive guidance is required, at least initially, to ensure that measurements are conducted correctly. The target population includes individuals not educated in this field, especially considering the project is directed towards women in the community.
- Repairability: If the device requires specific maintenance or is prone to malfunction, a local function is necessary to address such issues.

These parameters will help determine the suitability of a measurement device for the project, taking into account the local context, financial constraints, user capabilities, and the long-term goals of providing clean and safe drinking water to the community.

Product description

In this project, we focused on researching various existing pollutants, specifically addressing the issues that arose in previous studies conducted in Karla. The search for potential measurement tools to identify these pollutants was based on several parameters. Since our work was preliminary, a comprehensive prioritization of the parameters was not conducted. It is recommended that future teams continuing the project allocate appropriate weight to the various parameters in order to make an informed choice of the most suitable measurement tools. The main output presented in this chapter is a summary of our findings and guidance for the next team.

We conducted an extensive review of various measurement tools produced by different companies. There are many types of measurement tools, each with its own usage method. We focused on the "Dip" type of measurement tools, which are relatively inexpensive and straightforward to use. Some measurement tools are based on a paper strip or a tablet that is dipped in water. After the dip, the tester needs to remove the measurement tool from the water and wait for a certain period of time, which varies between different types of tests. The measurement tool changes its color according to the level of pollution in the water, and the tester can compare the color to the color chart provided with the test kit to understand the concentration of pollutants in the water.

Another type of measurement tool is a kit that contains a certain substance, whether in liquid or powder form, which needs to be added to a water sample to be examined. The tester is required to add a specified amount of the testing substance to the water, mix it, and wait until the color changes. Here too, the results are based on comparing the obtained color to a color chart. Additionally, there are electronic devices that can be placed in water, and the measurement result is displayed digitally on the device's screen.

The parameters we considered in choosing the measurement tools are as follows:

1. Ease of use - We rated the usability of each measurement tool as follows:
 - Score 1 is given to tools that require rapid dipping, providing an immediate result without waiting for more than a minute and requiring only dipping.
 - Score 2 is given to measurement tools that require adding a certain substance to the water, providing immediate results.
 - Score 3 is given to tools that require the tester to wait for a period of time exceeding a minute.

Most measurement tools that do not provide an immediate result within a few seconds require the tester to wait between an hour to 48 hours, during which the water sample must be kept sterile to prevent additional contamination that could affect the measurement results. We believe that for our target audience, this is a non-trivial task and therefore more challenging. Furthermore, electronic devices that require operations beyond dipping and color reading received a score of 3.

2. Impact of the pollutant on the individual consuming it - Some pollutants have an immediate impact on human health, while others, through prolonged consumption, may lead to long-term damage. It is noteworthy that for very high values, each of the pollutants can cause immediate harm and a variety of health problems. Therefore, the reference to pollutants that pose a long-term risk is only applicable in case of an abnormal and reasonably expected presence exceeding the allowable quantity.
3. Accuracy - The level of accuracy of the measurement tool and the range of values it can test. It is recommended to choose a measurement tool with sufficient accuracy to distinguish between extremely high values that indicate immediate danger and values that exceed the norm but do not constitute an immediate risk.
4. Price per unit - Some measurement tools are sold individually, while others come in packages of about 50-100 units. Some are for single use, and some are multi-use. To make an informed comparison, a single test of cost calculated.
5. Environmental consequences - As mentioned, some measurement tools are for single use, while others are multi-use. The reusable ones require the use of batteries, which need to be replaced at defined intervals. These batteries can contribute to environmental pollution if not disposed of properly. Single-use measurement tools are made of paper or sometimes plastic. The daily disposal of these tools after use can also negatively impact the environment. The environmental implications of each measurement tool need to be examined, and a reasonable solution for handling the waste from the tests must be considered.

From previous studies in which we examined water pollution specifically in Karla, consistent findings indicated the presence of several pollutants that affect both short-term and long-term aspects. The ten most common pollutants in Karla are as follows:

- pH - Values exceeding the allowable range may indicate water pollution. Water with a pH value below the minimum allowed can be too acidic, while values above the maximum allowed can be too alkaline. Prolonged consumption of water with pH values outside the norm can lead to health problems.

- E.coli - A biological pollutant that affects humans immediately and can cause various gastrointestinal issues.
- TDS (Total Dissolved Solids) - Indicates the level of dissolved minerals in the water. High TDS values may indicate the presence of excessive minerals in the water. Currently, there is no clear evidence of the impact of TDS on water quality.
- Iron - A long-term pollutant that can cause various diseases such as cancer and heart diseases.
- Coliform - A biological pollutant that affects humans immediately and can cause various gastrointestinal issues. However, the presence of coliform is not necessarily an indicator of pollution, as any contact of living organisms with the water can lead to its presence.
- Turbidity - A visual parameter that may indicate various pollutants, making it difficult to draw immediate or long-term conclusions.
- Nitrate - A sign of pollution from sources like runoff or manure. Prolonged consumption can lead to respiratory issues and other problems, especially in infants.
- Alkalinity - Not a parameter by itself, but monitoring changes in its value can indicate changes in the water source, such as the entry of muddy water.
- Residual chlorine - The presence of free chlorine in the water can affect taste and odor in the short term and cause additional issues in the long term.
- Fluorides - Prolonged consumption can cause fluorosis and dental problems.

A detailed breakdown of the various measurement tools that fulfill the criteria we selected and test for these ten pollutants, as well as additional pollutants, is provided in Appendix 3. Each measurement tool was analyzed based on the criteria mentioned above to determine the options for the next team to work on the project. We recommend conducting a thoughtful calculation that takes all these parameters into account in order to choose a single measurement tool or assemble a kit containing several measurement tools. All the above information refers to the testing aspect of the project. Ideally, the second part, which includes the application and the survey, as detailed in the discussion chapter, will be integrated with the testing to obtain a more comprehensive picture of the water quality and provide a platform that consolidates all the data.

Discussion

The project will focus initially on the state of Kerala, a sub-region in southwest India with a population of approximately 35 million (Government of Kerala, 2019). This project builds upon the initial steps taken in Kerala by a team from the Nitzan Lab at Tel Aviv University, which researches various dimensions of sustainable development in sub-Saharan Africa and Asia, addressing global poverty while preserving the environment and natural resources. The lab's team collaborates with the local Amrita University, serving as a bridge between the project and the community. The team worked in Kerala for several months, conducting surveys among locals to identify pollutants in drinking water and collecting water samples for laboratory testing to monitor pollutants. The research revealed a significant presence of various pollutants in the drinking water, highlighting the crucial need for daily water quality monitoring. In the future, the project aims to expand and pave the way for additional projects to enable water monitoring in communities where the need exists.

Throughout the work, we were exposed to the complexity involved in providing effective water monitoring in developing countries. There are several challenges, with the primary difficulty being conducting on-site water tests. Maintaining testing equipment under the required conditions is not a simple task, even for an experienced team, let alone for individuals in villages or cities who may not be accustomed to it and might not fully grasp the importance of adhering to proper protocols. Additionally, laboratory water testing does not offer efficient and rapid solutions for household use in India (World Health Organization, 2012). Therefore, we focused on various tools and articles related to measuring different pollutants, primarily those that are user-friendly, allowing for quick and simple solutions to water purification challenges.

Regarding recommendations for the project's continuation and implementation in the field, it is necessary to adapt the WQI (Water Quality Index) based on the list of pollutants relevant to the pilot location. Subsequently, a monitoring kit needs to be constructed, with a smart choice of measurement tools as described in the product description section. Calculating the cost-benefit ratio should consider parameters such as price, pollutant risk level, environmental impact, and more. The goal is to create an affordable, sustainable, user-friendly, and accurate kit suitable as an initial MVP (Minimum Viable Product).

The primary goal of the project is to gather data on the water situation in various communities. As part of this effort, the team from the Nitzan Lab at Tel Aviv University conducted a survey last

year, where participants reported on the water they drink – its source, color, taste, odor, and more. The survey questions need to be refined to provide daily-based responses, both in terms of length and content, aligned with the criteria measured for water quality based on the WQI (Water Quality Index). The survey responses can indicate the presence of pollutants. The survey will be divided into two types: one conducted periodically every few months or after significant climatic events, discussing consumption habits and general water condition, and the other conducted on a daily or weekly basis, pinpointing the specific water condition at a given point in time.

To distribute the survey, an option to create an app was proposed, one that doesn't require continuous internet usage or through a booklet, recognizing that even as mobile usage increases, continuous internet connectivity is not reliable in many developing countries. However, the need to collect data from the field to the lab should be considered, and the platform should be adapted accordingly.

With the user interface, individuals can assess whether their water is suitable for consumption based on survey responses and measurement data. The specific questions asked, selected from a general question repository, depend on each response, leading to different follow-up questions based on suspicion. The integration of the survey with the interface will serve as a kind of water quality guide. Initially, the identification of pollutants present in the water (chemical, biological, solid, floating, etc.) will be done. The interface, knowing the acceptable ranges for each pollutant, will provide an answer – whether the water is approved for use or not. Identification of viruses and bacteria that can cause immediate issues such as diarrhea and other gastrointestinal problems will lead to negative results. For chemical pollutants like metals, causing long-term harm, the app will store data to signal the need for future treatment. As a result, based on the survey outcomes, recommendations can be made for appropriate water treatment methods that users can apply themselves (up to a certain level) for suspected water sources.

The proposed economic model will be B2B based, meaning the revenue will come from businesses that advertise on the app or promote their devices. Partnerships with filtration device manufacturers can also be established, offering them "advertising" on the app in exchange for discounts to app users and a percentage of sales revenue for us. This would create a new commerce cycle. Suppliers will offer their devices, and users will purchase them based on the results from the survey and app recommendations. If the pilot is successful, this model has the potential to become more global. The goal is for the app to be free. Further details on the business model can be found in Annex 7

Bibliography

- Abdi, M. Q., & Delavari, E. (2018). *Adaptive Water Management Concepts, Principles and Applications for Sustainable Development*. Springer International Publishing.
- Allen, M. J., Taylor, R. H., & Geldreich, E. E. (1980). The occurrence of microorganisms in water main encrustations. *American Water Works Association*, 72(11), 614-625.
- Ananth, M., Rajesh, R., Amjith, R., Achu, A. L., Valampampil, M. J., Harikrishnan, M., . . . Nujum, Z. T. (2018). Contamination of Household Open Wells in an Urban Area of Trivandrum, Kerala State, India: A Spatial Analysis of Health Risk Using Geographic Information System. *Environmental Health Insights*, 12(1).
- Anoop, C. K., Arathy, A., Markose, M., Merin, M., & Riya, B. (2019). Comparative Study of WQI - Kerala Region. *International Journal of Scientific Research and Review*, 8(6), 335-339.
- Castillo, G., Duarte, R., Ruiz, Z., Marucic, M. T., Honorato, B., Mercado, R., . . . Dutka, B. J. (1994). Valuation of Disinfected and Untreated Drinking Water Supplies in Chile by the H₂S Paper Strip Test. *Department of Civil Engineering, University of Chile*, 28(8), 1765-1770.
- Davis, J', Pickering, A' J', Rogers, K', Mamuya, S & 'Boehm, A' B .(2011) .'The Effects of Informational Interventions on Household Water Management, Hygiene Behaviors, Stored Drinking Water Quality, and Hand Contamination in Peri-Urban Tanzania .*The American Society of Tropical Medicine and Hygiene*.184-191 ,
- Government of Kerala, K. S. (2019). *Economic Review 2018*. Thiruvananthapuram, Kerala, India.
- Hamidreza, A., & Zohreh, A. (2012). Evaluation of the Relation between Water Pollution and Economic Growth in Developing Countries and Developed Countries. *Journal of Iranian Energy Economics*, 3(1), 1-16.
- Karthick, B., Boominathan, M., & Ramachandra, T. V. (2015). Evaluation of the Quality of Drinking Water in Kerala State, India. *Centre for Ecological Sciences, Indian Institute of Science*, 203-218.

Kathuria, V. (2006). Controlling water pollution in developing and transition countries - lessons from three successful cases. *Journal of Environmental Management*, 78(4), 405–426.

Lakshmi, E., & Madhu, G. (2014). An Assessment of Water Quality in River Periyar, Kerala, South India Using Water Quality Index. *Journal of Environmental Science, Toxicology and Food Technology*, 8(8), 2319-2399.

Manja, K. S., Maurya, M. S., & Rao, K. M. (1982). A Simple Field Test for the Detection of Faecal Pollution in Drinking Water. *Bulletin of the World Health Organization*, 60(5), 797-801.

Okyere, C' Y', Pangaribowo, E' H', Asante, F' A & ,Braun, J' v .(2017) .'The Impacts of Household Water Quality Testing and Information on Safe Water Behaviors: Evidence from a Randomized Experiment in Ghana .*ZEF- Center for Development Research, University of Bonn*. 5319-5329.

Sustainable Development Goal Website.
<https://sustainabledevelopment.un.org/?menu=130> [accessed: 25/05/2020].

Trent, M', Dreibelbis, R', Bir, A', Tripathi, S' N', Labhasetwar, P', Nagarnaik, P . . . ,Brown, J .(2018) .'Access to Household Water Quality Information Leads to Safer .
Environmental Science and Technology

Sadan, A. (2002). Empowerment and Community Planning: Theory and Practice of Social and Human Solutions. Hakibbutz Hameuchad..

Appendix

• Appendix 1 - Water Quality Testing Table in Israel • Appendix 2 - Pollution Table • Appendix 3 - Measurement Instruments Table • Appendix 4 - Similar Studies Table • Appendix 5 - Discussion Tables • Appendix 6 - Research on the Prayagraj River, India • Appendix 7 - Business Canvas

Appendix 1 - Prices for Water Quality Testing in Private Property according to Mekorot - Regional Water Corporation, as of June 2020. Can be found at:

<https://meitav-pt.co.il/RichText/GeneralPage.aspx?nodeId=538>

עלות (ללא מע"מ)	שם הבדיקה
₪ 461.01	בדיקת מתכות
₪ 320.70	בדיקת מזהמים בקטריאליים, עכירות ועקבות חומר חיטוי פעיל
₪ 571.25	בדיקת מתכות, מזהמים בקטריאליים, עכירות ועקבות חומר חיטוי פעיל

Appendix 5 - Tables for Discussion

Table 1:

TABLE 1: Brief description of sampling sites

Sample No	Sampling Points
S1	Edamula, upstream of the river
S2	Pathalam Bund North (Edayar), before reaching the bund
S5	Pathalam Bund South (Edayar), after the bund
S8	Sudchemi (Edayar), close to the sudchemi factory in the edayar region
S11	Pallikadavu (Edayar), near the bank of edayar region with few small-scale industries
S14	Kuzhikandam, discharge point of industries like HIL, Merchem
S15	Eloor Ferry Right, confluence point of the two branches to main river.
S16	Eloor Ferry Left, confluence point of two branches to the main river.

Table 2:

H₂S paper strip and water supplies

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Table 1. Type and number of samples: summary of study

Geographical region	Well (ground/draw)	Untreated				Disinfected			Total
		Fountain/river	Watershed	Trapped fog	Standpipe	Chlorinated (tap)	Tank truck	Barrel	
IV	54	27	—	9	18	90	6	3	207
RM	15	22	—	—	8	111	—	44	200
IX	15	19	58	—	—	123	—	—	215
Total	84	68	58	9	26	324	6	47	622

Table 3:

שם המזהם	טיפול
PH, Conductivity, Total Dissolved Solids	digital meters
Total hardness and calcium	Ethylene Di-amine tetra acetic acid (EDTA)
chloride	Argentometric method
Bio-chemical oxygen demand and dissolved oxygen	Winkler methods
Iron	spectrophotometrically with O – phenanthroline reagent
sulphate	Turbidimetric method with barium chloride reagent
Sodium and potassium	flame photometer
Nitrate-N	Brucine- sulphanilic acid
magnesium	Gravimetric method
fluoride	SPADNS method
Phenol	mino-antipyrène and chloroform extraction method

Table 4:

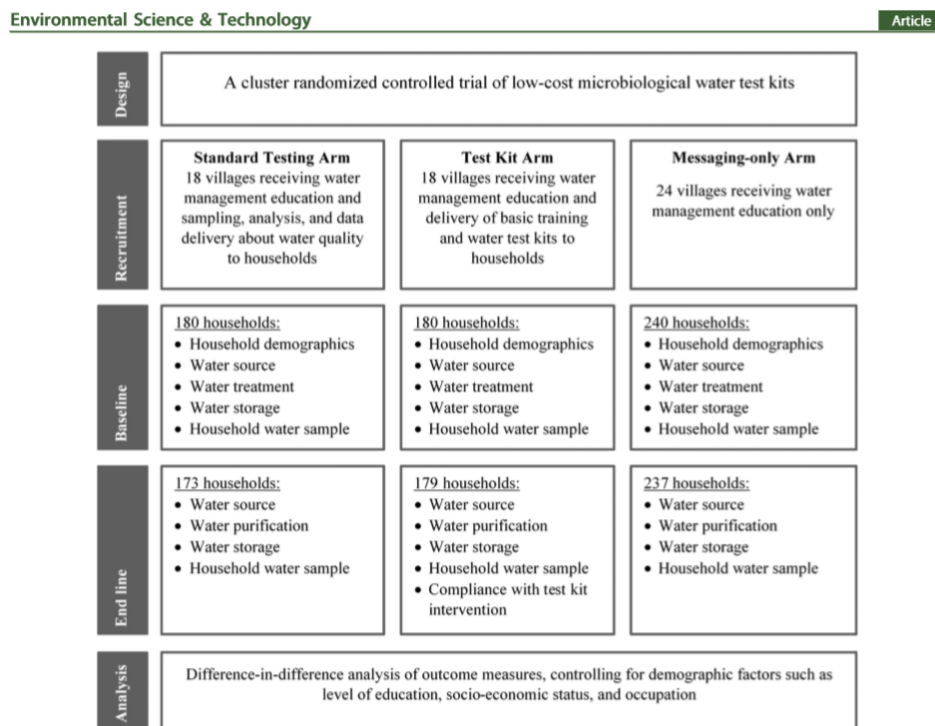


Figure 1. CONSORT³⁹ diagram describing the cluster randomized controlled trial design of the study.

Table 5:

Table 2. Differences in Water Quality and Key Behaviors between Treatment Cohorts and Messaging-Only Group

	baseline	end line ¹	DiD ² (95% CI)	p-value
Mean Log ₁₀ <i>E. coli</i> cfu/100 mL of Household Drinking Water (SD)				
standard testing arm	1.38 (1.57)	0.89 (1.27)	−0.93 (−1.28, −0.58)	<0.01
test kit arm	1.09 (1.54)	0.65 (1.07)	−0.89 (−1.14, −0.64)	<0.01
messaging-only arm	1.42 (1.76)	1.87 (1.55)	(referent)	
Proportion of Households Reporting Boiling Drinking Water Prior to Use in Previous Two Weeks (SE)				
standard testing arm	0.03 (0.01)	0.45 (0.04)	0.38 (0.27, 0.48)	<0.01
test kit arm	0.02 (0.01)	0.34 (0.04)	0.28 (0.18, 0.39)	<0.01
messaging-only arm	<0.01 (<0.01)	0.04 (0.01)	(Referent)	
Proportion of Households Reporting Using a Commercial Water Filter in Previous Two Weeks (SE)				
standard testing arm	0.01 (0.01)	0.01 (0.01)	0.00 (−0.01, 0.00)	0.32
test kit arm	0.02 (0.01)	<0.01 (<0.01)	−0.02 (−0.04, 0.01)	0.21
messaging-only arm	<0.01 (<0.01)	<0.01 (<0.01)	(referent)	
Proportion of Households Using a Cover or Lid on Their Water Storage Container (SE)				
standard testing arm	0.96 (0.01)	0.98 (0.01)	−0.03 (−0.08, 0.02)	0.26
test kit arm	0.93 (0.02)	1.00 (<0.01)	0.02 (−0.08, 0.12)	0.66
messaging-only arm	0.96 (0.01)	1.00 (<0.01)	(referent)	
Proportion of Households with Soap Available at Handwashing Station at Time of Survey (SE)				
standard testing arm	0.94 (0.02)	0.97 (0.01)	−0.05 (−0.12, 0.02)	0.17
test kit arm	0.89 (0.02)	0.99 (0.01)	0.02 (−0.08, 0.13)	0.67
messaging-only arm	0.92 (0.02)	1.00 (<0.01)	(referent)	
Proportion of Households with at Least One Case of Diarrhea in the Previous Seven Days (SE)				
standard testing arm	0.12 (0.02)	0.02 (0.01)	−0.02 (−0.09, 0.05)	0.54
test kit arm	0.07 (0.02)	0.01 (0.01)	0.02 (−0.05, 0.10)	0.51
messaging-only arm	0.08 (0.02)	0.01 (0.01)	(referent)	

¹End line visits were conducted approximately 4 weeks after the initial baseline visit. ²Difference-in-difference estimator relative to messaging-only arm, adjusted for baseline differences in education level completed and below poverty line status. ³Baseline was *n* = 173, end line was *n* = 173.

⁴Baseline was *n* = 178, end line was *n* = 179. ⁵Baseline was *n* = 233, end line was *n* = 233.

Appendix 6 - Study on the River Prair

An additional document that describes the water quality index of the Prair River, India, using parameters such as pH, conductivity, Total dissolved solids, Total hardness, Dissolved Oxygen, Chemical Oxygen Demand, Calcium, Magnesium, Sulphates, Chlorides, Nitrate-N, Iron, Fluoride, and Phenol. The samples were collected from April 2007 to May 2008, each time in the morning hours. Pollutants were measured using various methods (see Appendix 5, Table 3). The research objectives were to depict the pollution status of the river using the water quality index and to analyze the relationship between various parameters using the SPSS statistical software. Understanding the correlation between parameters and water quality index. The comprehensive water quality index indicated a "poor" water quality in the river. Regression analysis showed that the water quality index decreases with an increase in the concentration of parameters and vice versa for parameters such as calcium, sulphate, chloride, nitrate-nitrogen, total hardness, fluoride, and conductivity. The analysis found that the water quality index increases with an increase in pH and dissolved oxygen. The results indicate a significant influence on the water quality index when multiple parameters deviate from the norm, but if each parameter deviates individually from the standard, there is no significant effect on the index. (Lakshmi & Madhu, 2014)._____

Appendix 7 - Business Proposal

We want to address the economic aspect of the project, with a focus on the primary research location: the Karla district in southern India. Given the current economic situation in India, the majority of the population earns less than \$2 per day. The question that arises is how to generate purchasing power within this population. We see tremendous potential for a product of this kind within such a large population, as there is a broad market in India, but the accessibility of the product needs to be different and affordable.

To implement the project, funding is needed. One option is to seek funding from the Karla government, investors who see business potential in the venture, or philanthropic organizations. Initially, the funds will be directed towards developing the application, employee salaries to assist with user training, the cost of required servers, customer service, and marketing (distribution in India). Cost reduction is possible by exploring the option of developing the application through Amrita University.

The marketing and distribution of the product are significant, as most of the population in India lacks the ability or sufficient motivation to purchase water purification devices. Therefore, one of the goals of the project is to increase water awareness among women in India. Once the entire population in Karla understands the importance of the idea, the project can be extended to other locations.

The aspiration is for the application to be free. Collaboration with companies that manufacture or sell filtration devices is possible. This will be a platform that connects suppliers and customers. The general idea is that women in India will take a survey to receive an indication for a specific filtration device, and the application will automatically direct them to the relevant device page, where various types of devices will be listed with quality and cost ratings.

Several economic models can be considered for the project:

1. Creating competition among suppliers: After completing the survey, women will be directed to a page with recommendations for relevant devices. Each supplier will receive a rating based on user recommendations. Incentives for suppliers to set lower prices can be established.
2. Funding: Suppliers can sell products at a pre-determined price, and a portion of the product cost will be funded by a specific organization (government or philanthropic). This way, people can afford to purchase the product at a reduced price. For example, they could enter a coupon code to receive a discount.
3. Investment: If the project is funded by an investor, a business plan must be developed. How will the application generate revenue?
 - Advertising revenue: Advertisements can appear within the application, and income can be generated from them.
 - Commission: The website will operate in a B2B2C model, acting as a mediator between the supplier and the customer. A small commission fee will be charged for each transaction, similar to Amazon or eBay. However, in this case, it is essential to reduce the product cost, not inflate it.

If the pilot is successful, this model can be expanded globally, serving not only India but also other developing countries. A central question that arises is how to motivate people to buy these devices. What interests the customer is not the solution but the problem they have. Once

women are water-aware in Karla, they will be motivated to invest in solving this problem. Therefore, within the project, it would be prudent to invest in marketing, particularly television advertisements, as statistics show that most people in India have TVs.

Another option is to create a community where locals can develop businesses and sell devices. When it comes from within the population, it becomes easier to distribute the product. Therefore, we want to present -Lean Canvas⁸. This is a business plan. A business plan defines the initiative, centralizes all relevant information, assists in strategic decision-making, dictates what is needed in the project at each stage, provides an internal perspective, and also offers information about the external market.

PROBLEM Monitoring and identification of safe drinking water in developing areas, women's empowerment, and creating gender equality are crucial for the woman's role in household sustainability. SDG Goal 6 - Clean Water and Sanitation	SOLUTION Water purification measurement tools (devices)	UNIQUE VALUE PROPOSITION Clean water Improved quality of life Sustainability	UNFAIR ADVANTAGE In India, the topic of water quality is relatively new	CUSTOMER SEGMENTS Household, primarily women as they take care of drinking water daily. Early Adopters: People who consider water quality important and want to improve their quality of life
	KEY METRICS Improvement in the lives of millions of people.		CHANNELS Television commercials, government advertisements, various events	
COST STRUCTURE Starting a business, purchasing measuring equipment, hiring			REVENUE STREAMS Revenue from sales of measuring equipment	

⁸ Lean Canvas is an adaptation of Business Model Canvas